

PROJECT REPORT

NovaCardio: Heart Failure Detection & Management System

Submitted to



Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal

In partial fulfillment of the requirements for the
Bachelor of Technology in
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Submitted By

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Enrollment No: 0902IT221017

Under the Guidance of

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Certificate

This is to certify that the project report entitled “**NovaCardio: Heart Failure Detection & Management System**” is a bona fide work carried out by **Ayush Morbar** (Enrollment No: 0902IT221017) under my supervision and guidance. This project is submitted to **Rajiv Gandhi Proudhyogiki Vishwavidyalaya, Bhopal** in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in **Information Technology**, during the academic session **January–June 2026**.

The project embodies the original work done by the student. To the best of my knowledge, the matter embodied in this report has not been submitted to any other University or Institution for the award of any degree or diploma. The student has demonstrated exceptional dedication, engineering logic, and technical rigor in developing this cloud-deployed Clinical Decision Support System.

Prof. J.S. Pahariya

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Declaration

I, **Ayush Morbar**, hereby declare that the project titled “**NovaCardio: Heart Failure Detection & Management System**” submitted to Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal, is a record of an original work done by me under the guidance of **Prof. J.S. Pahariya**.

The empirical findings, source code, data preprocessing pipelines, and clinical mapping systems presented in this report have been entirely engineered by me and have not been submitted elsewhere for the award of any other degree, diploma, fellowship or other similar titles. All resources, datasets, and literature utilized in this project have been acknowledged and cited in the bibliography.

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I would like to extend my thanks to the open-source community, particularly the developers of Streamlit, CatBoost, and SHAP, whose modern frameworks made the cloud deployment of this system possible.

Finally, I dedicate this work to my parents, family, and friends. Their unwavering moral support, patience, and motivation kept me focused and driven throughout the grueling phases of development, debugging, and compilation.

Ayush Morbar

Abstract

Cardiovascular diseases remain the leading cause of global mortality. Although standard machine learning models offer high accuracy in disease detection, they treat medical diagnostics as a “black box,” relying on raw accuracy metrics without providing actionable clinical insights. This project engineers **NovaHeart**, a robust end-to-end Clinical Decision Support System (CDSS) designed to bridge the gap between predictive modeling and automated patient management.

The detection engine stratifies cardiovascular risk using a Gradient Boosted algorithm (CatBoost), optimized via Optuna for categorical tabular datasets. To ensure structural data integrity, the pipeline implements defensive K-Nearest Neighbors (KNN) imputation strictly fitted on training sets to prevent data leakage, alongside the Synthetic Minority Over-sampling Technique (SMOTE-NC) to safely balance mixed-type clinical cohorts.

To ensure maximum clinical safety, the system bypasses default probability thresholds. Using the analysis of the Precision-Recall curve, a custom operational threshold of 0.5723 was identified, strictly guaranteeing a 95% Recall rate to minimize fatal False Negatives. Furthermore, the system integrates Explainable AI (SHAP) to map predictions back to physiological biomarkers, establishing clinical trust. Ultimately, NovaHeart feeds the ML inferences into a deterministic medical rule-engine based on Ejection Fraction (EF%), fluid retention, and functional staging of NYHA. The entire architecture is successfully operationalized and deployed in the cloud as a live interactive Streamlit Point-of-Care dashboard, transforming predictive mathematics into accessible healthcare technology.

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List of Abbreviations

AI Artificial Intelligence

ML Machine Learning

HF Heart Failure

ECG Electrocardiogram

CDSS Clinical Decision Support System

EHR Electronic Health Record

SHAP Shapley Additive exPlanations

SMOTE-NC Synthetic Minority Over-sampling Technique for Nominal and Continuous

NYHA New York Heart Association

EF Ejection Fraction

KNN K-Nearest Neighbors

TPE Tree-structured Parzen Estimator

GUI Graphical User Interface

Chapter 1

Introduction

1.1 Background and Motivation

Cardiovascular disease (CVD) remains the most formidable global health challenge of the 21st century, accounting for an estimated 17.9 million deaths annually according to the World Health Organization (WHO) [1]. Among the various manifestations of CVD, Heart Failure (HF)—a chronic, progressive condition in which the heart muscle is unable to pump enough blood to meet the body’s needs for blood and oxygen—places an unprecedented burden on healthcare systems. The diagnosis and staging of heart failure require the continuous monitoring of a wide array of physiological biomarkers, hemodynamic parameters, and patient symptomatology.

Modern cardiology relies heavily on Electronic Health Records (EHRs) and complex diagnostic tests, ranging from resting Blood Pressure (BP) and Fasting Blood Sugar (FBS) to advanced echocardiographic metrics like Ejection Fraction (EF) and Electrocardiogram (ECG) wave abnormalities. The sheer volume, high dimensionality, and complex non-linear interactions of these tabular data make it exceedingly difficult for human practitioners to rapidly synthesize a risk profile and define an immediate, error-free treatment plan during emergency triage.

In recent years, the intersection of Artificial Intelligence (AI) and Machine Learning (ML) has shown tremendous promise in automating pattern recognition within clinical datasets. However, the adoption of ML in actual hospital settings has been severely hampered. Medical professionals cannot ethically or legally prescribe life-altering medications based solely on a computer that outputs a probability score. They require transparency, rigorous data validation, and alignment with established clinical protocols. This project NovaCardio: Heart Failure Detection & Management System, is motivated by the urgent need to bridge the chasm between raw predictive mathematics and deployable, trustworthy clinical management.

1.2 Problem Statement

Despite the proliferation of machine learning models in the academic literature, standard diagnostic pipelines suffer from three critical architectural flaws that prevent their deployment in real-world healthcare:

1. **Fragility to Real-World Data Anomalies:** Clinical data is inherently messy. Models deployed without defensive data schemas often crash when faced with missing inputs (e.g., a skipped blood test) or categorical imbalances, rendering them useless in fast-paced hospital environments.
2. **The “Black-Box” Interpretability Crisis:** High-performing models like Deep Neural Networks and Gradient Boosting Machines offer excellent predictive accuracy but lack biological context. A model predicting a risk of 98% of cardiac arrest is clinically untrustworthy if the attending physician cannot see the underlying variables driving that prediction.
3. **The Actionability Gap:** Providing a physician with a raw risk probability does not instruct them on the subsequent medical intervention. Current models do not integrate with deterministic treatment guidelines (such as ACC/AHA protocols), leaving a gap between diagnosis and treatment.

1.3 Objectives of the Study

To resolve these systemic flaws and build a production-ready Clinical Decision Support System (CDSS), the primary objectives of this project are the following:

- **Defensive Data Engineering:** Implement strict feature schema-locking and leak-free K-Nearest Neighbors (KNN) imputation to handle incomplete patient records dynamically without compromising mathematical integrity.
- **High-Recall Algorithmic Optimization:** Train a CatBoost gradient boosting classifier utilizing Bayesian hyperparameter optimization (Optuna). Crucially, the objective is to apply Precision-Recall curve analysis to manually tune the decision threshold, strictly guaranteeing a 95% Sensitivity (Recall) rate to minimize fatal False Negatives.
- **System Transparency via Explainable AI:** Deploy Shapley Additive exPlanations (SHAP) to decode the model’s inner logic, generating global physiological feature summaries and local, patient-specific waterfall charts.
- **Cloud Deployment & Clinical Translation:** Develop a deterministic medical rule engine that maps the probability of ML-derived risk alongside key biological markers (e.g., Ejection Fraction, NYHA class) to produce standardized Guideline-Directed Medical Therapy (GDMT) care plans, fully hosted and deployed via a live Streamlit web application.

1.4 Scope of the Project

The scope of this project encompasses the development, training, optimization, and serialization of an MLOps pipeline. It uses the highly dense Z-Alizadeh Sani (CAD) dataset, discarding standard but limited datasets (e.g., the UCI Cleveland dataset) after a structural audit [2]. The project focuses strictly on tabular clinical data and culminates in the export of production-ready artifacts (CatBoost .cbm binaries, pickled imputers, and JSON schema files). These artifacts are explicitly integrated into a web-based Python dashboard (Streamlit) for the use of real-world clinicians.

1.5 Organization of the Report

The remainder of this report is organized as follows: **Chapter 2** provides a comprehensive review of the literature on ML in cardiology and explainable AI. **Chapter 3** discusses the theoretical and mathematical background of the algorithms utilized, including CatBoost, SMOTE-NC, and SHAP. **Chapter 4** details the system architecture, data dictionaries, requirements analysis, and dataset audits. **Chapter 5** covers the Python implementation details, highlighting defensive programming paradigms, software testing, and the Web UI. **Chapter 6** presents the evaluation metrics, threshold optimization results, failure mode analysis, and visual SHAP interpretations. Finally, **Chapter 7** concludes the study, discussing ethical considerations, societal impact, and avenues for future scope and deployment.

Chapter 2

Literature Review

2.1 Traditional Statistical Methods in Cardiology

The pursuit of quantifying cardiovascular risk predates the modern era of machine learning. The Framingham Heart Study, initiated in 1948, yielded the Framingham Risk Score (FRS), a sex-specific algorithm used to estimate the 10-year cardiovascular risk of an individual [3]. Although foundational, traditional statistical models such as Logistic Regression and Cox Proportional Hazards models rely heavily on the assumption of linearity between independent variables and the logarithmic odds of the dependent variable. However, medical data are highly non-linear [4]. The interaction between fasting blood sugar, age, and left ventricular hypertrophy in a patient cannot be perfectly captured by linear coefficients, prompting a shift toward algorithmic machine learning.

2.2 Machine Learning and Ensemble Methods

Over the past decade, the literature has demonstrated the vast superiority of ensemble tree-based methods over traditional algorithms in tabular medical datasets. Random Forests [5] introduced bagging to reduce variance, while Gradient Boosting Machines (GBMs) introduced boosting—building sequential trees that correct the residual errors of their predecessors [6].

Recent advances have led to specialized GBM implementations such as XGBoost, LightGBM, and CatBoost. CatBoost (categorical boost), developed by Prokhorenkova et al. (2018) [7], has proven to be exceptionally effective in clinical informatics. Unlike XGBoost, which requires extensive one-hot encoding that dilutes feature space, CatBoost handles categorical variables natively using target-based statistics. Furthermore, its use of oblivious decision trees acts as a powerful regularizer, mitigating the severe overfitting that plagues deep neural networks when trained on small-to-medium clinical cohorts (e.g., $N < 1000$).

2.3 Handling Class Imbalance and Missing Data

A ubiquitous challenge in medical informatics is class imbalance; natural datasets inherently contain more negative (healthy) cases than positive (diseased) cases. Training on skewed data biases the classifier toward the majority class, destroying Recall. Chawla et al. (2002) introduced the SMOTE (Synthetic Minority Over-sampling Technique) to address this [8]. However, standard SMOTE utilizes linear interpolation, which fails when applied to nominal (categorical) data—often synthesizing impossible clinical values (e.g., a “Smoking Status” of 0.4). This project relies on SMOTE-NC, an extension of SMOTE specifically designed to handle mixed continuous and nominal datasets safely, penalizing synthetic points that violate categorical boundaries.

Defensive data engineering also requires robust imputation. Although many studies utilize median/mean imputation, this artificially compresses dataset variance. The K-Nearest Neighbors (KNN) imputation has emerged in the literature as a superior method, reconstructing missing physiological variables by calculating the Euclidean proximity of surrounding patient profiles [9].

2.4 Explainable AI (XAI) in Healthcare

As ML models have grown in complexity, regulatory bodies (such as those enforcing GDPR and HIPAA) have emphasized the “Right to Explanation” in algorithmic decision-making [10]. LIME (Local Interpretable Model-agnostic Explanations) was an early attempt to approximate black-box models locally [11].

However, Lundberg and Lee (2017) revolutionized the field of XAI by introducing SHAP (Shapley Additive exPlanations) [12]. Based on cooperative game theory, SHAP calculates the exact marginal contribution of every feature to the final prediction, ensuring both local accuracy and global consistency. The integration of SHAP transforms an opaque GBM into a transparent diagnostic tool, perfectly aligning with the rigorous demands of medical professionals.

2.5 Clinical Decision Support Systems (CDSS)

While predictive models are common in computer science literature, true Clinical Decision Support Systems (CDSS) are rare. A CDSS must bridge the gap between statistical probability and clinical action. The American College of Cardiology (ACC) and the American Heart Association (AHA) define strict treatment pathways for heart failure [13]. For example, a patient with an Ejection Fraction (EF) $< 40\%$ is classified as having Heart Failure with Reduced Ejection Fraction (HFrEF) and requires Guideline-Directed Medical Therapy (GDMT) such as Beta-blockers or ARNI.

2.6 Summary of Literature Gap

The primary gap identified in the current literature is the disconnect between isolated ML experiments and clinical realities. Studies frequently report 90%+ accuracies but do not provide a mechanism to handle missing input data, do not explain biologically mathematical inferences, and do not produce a care plan that complies with GDMT. NovaCardio: Heart Failure Detection & Management System is specifically engineered to fill this gap, unifying advanced data balance, XAI, and deterministic medical guidelines into a single deployable pipeline.

Chapter 3

Theoretical Background and Algorithmic Framework

This chapter details the mathematical foundations of the algorithms and data engineering techniques utilized within the NovaHeart pipeline. A rigorous understanding of these mechanics is essential for validating the system’s clinical reliability.

3.1 K-Nearest Neighbors (KNN) Imputation

Clinical datasets often suffer from missing values (NaNs) due to equipment failure or skipped diagnostic tests. Simple imputation methods, such as mean or median substitution, artificially compress the variance of the dataset and can distort multi-dimensional clinical relationships.

The K-Nearest Neighbors (KNN) imputation algorithm resolves this by operating in a multi-dimensional feature space. Let a patient record be represented as a vector $x \in \mathbb{R}^d$. If a feature x_j is missing for a target patient, the algorithm computes the distance between the target patient and all other patients in the training set using the Euclidean distance metric for all observed features:

$$d(x, y) = \sqrt{\sum_{i \in \text{Observed}} (x_i - y_i)^2} \quad (3.1)$$

The algorithm then selects the k patients with the smallest distance (the “nearest neighbors”). The missing value x_j is imputed by calculating a weighted average of the j -th feature of these neighbors k , where the weights are typically inversely proportional to the distance:

$$\hat{x}_j = \frac{\sum_{m=1}^k w_m y_{mj}}{\sum_{m=1}^k w_m}, \quad \text{where } w_m = \frac{1}{d(x, y_m)^2} \quad (3.2)$$

3.2 SMOTE-NC (Synthetic Minority Over-sampling Technique)

The Z-Alizadeh Sani dataset, like most medical cohorts, exhibits a class imbalance. Training on imbalanced data causes gradient boosting algorithms to prioritize the global loss function by simply predicting the majority class, leading to high Accuracy but catastrophic False Negative rates (low Recall).

Although the standard SMOTE algorithm effectively synthesizes continuous data, it fails on nominal (categorical) data. SMOTE-NC overcomes this by introducing a modified distance metric. For continuous features, it calculates the Euclidean distance. For nominal features, it assigns a penalty based on the median of standard deviations of all continuous features for the minority class.

If x_1 and x_2 are two instances in the minority class, the SMOTE-NC distance $D(x_1, x_2)$ is the following:

$$D(x_1, x_2) = \sqrt{\sum_{i=1}^C (x_{1i} - x_{2i})^2 + \sum_{j=1}^N \text{Penalty}(x_{1j}, x_{2j})^2} \quad (3.3)$$

where C is the set of continuous features, N is the set of nominal features, and the penalty is 0 if the nominal categories match, and a fixed constant (median standard deviation) if they differ. Synthetic points are then generated along the line segment joining the continuous features of the k -nearest minority neighbors, while the nominal feature is determined by the majority vote of those neighbors.

3.3 CatBoost (Categorical Boosting)

CatBoost, an implementation of Gradient Boosting Decision Trees (GBDT), was selected as the core detection engine due to its algorithmic superiority in handling tabular data and its resistance to overfitting.

3.3.1 Gradient Boosting Mathematics

GBDT is an ensemble technique that builds trees sequentially. Let the dataset be $D = \{(x_i, y_i)\}_{i=1}^n$. The goal is to find a function $F(x)$ that minimizes the expected value of a loss function $L(y, F(x))$.

In iteration t , the algorithm computes the negative gradient of the loss function evaluated in the predictions of the current model:

$$g_i^{(t)} = - \left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)} \right]_{F(x)=F_{t-1}(x)} \quad (3.4)$$

Then a new decision tree $h_t(x)$ is trained to predict these gradients $g_i^{(t)}$. The model is updated as:

$$F_t(x) = F_{t-1}(x) + \eta h_t(x) \quad (3.5)$$

where η is the learning rate.

3.3.2 Target-Based Categorical Encoding

Traditional GBMs require One-Hot Encoding for categorical variables, which exponentially increases dimensionality and creates sparse matrices. CatBoost solves this using Target Statistics (TS). For a categorical feature x_k , its value for the i -th instance is replaced by the expected target value, calculated dynamically to prevent target leakage:

$$\hat{x}_{i,k} = \frac{\sum_{j=1}^{i-1} [x_{j,k} = x_{i,k}] \cdot y_j + a \cdot p}{\sum_{j=1}^{i-1} [x_{j,k} = x_{i,k}] + a} \quad (3.6)$$

where a is a prior weight and p is the prior probability.

3.3.3 Oblivious Decision Trees

CatBoost utilizes symmetric (oblivious) decision trees. In an oblivious tree, the same splitting criterion is used across all nodes at the same depth level. This symmetry ensures that the tree is strictly balanced, which dramatically reduces prediction time (critical for live CDSS APIs) and acts as a strong regularizer against overfitting on small clinical datasets.

3.4 Optuna Bayesian Hyperparameter Optimization

Manual tuning of a CatBoost model (e.g., Grid Search) over an infinite continuous space (like learning rate or L2 regularization) is mathematically inefficient. Optuna resolves this using Bayesian Optimization, specifically the Tree-structured Parzen Estimator (TPE).

TPE models the probability of the hyperparameters given a target score (e.g., Recall). It defines two probability density functions (PDFs) based on a threshold y^* (which separates the top performing trials from the rest):

$$p(x|y) = \begin{cases} l(x) & \text{if } y < y^* \text{ (Better performance)} \\ g(x) & \text{if } y \geq y^* \text{ (Worse performance)} \end{cases} \quad (3.7)$$

The algorithm then selects the next set of hyperparameters x that maximizes the Expected Improvement (EI), which is directly proportional to the ratio:

$$\text{EI}(x) \propto \frac{l(x)}{g(x)} \quad (3.8)$$

This ensures that the algorithm spends its computational budget exploring highly promising regions of the hyperparameter space.

3.5 Explainable AI: Shapley Additive exPlanations (SHAP)

A gradient boosted ensemble composed of hundreds of oblivious trees is completely uninterpretable to a human physician. SHAP, rooted in cooperative game theory, deconstructs this black box.

In game theory, the Shapley value determines the fair distribution of a payout among players in a coalition. In the context of NovaHeart, the “game” is the prediction task, the “payout” is the difference between the actual prediction and the base (average) prediction, and the “players” are the clinical features (e.g., Age, EF%, BP).

The SHAP value ϕ_i for a feature i represents its marginal contribution to the prediction, averaged over all possible permutations of feature coalitions:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{i\}) - f_x(S)] \quad (3.9)$$

This additive property guaranties that the sum of the SHAP values for all features equals the difference between the model’s output for the specific patient and the global expected output.

$$f(x) = \phi_0 + \sum_{i=1}^M \phi_i \quad (3.10)$$

where ϕ_0 is the expected base value. This mathematical guaranty is what allows the generation of local Waterfall plots, visually proving to a clinician exactly how the model arrived at its conclusion.

Chapter 4

System Architecture and Methodology

The NovaHeart Clinical Decision Support System (CDSS) is designed as a modular, high-reliability, machine learning operations (MLOps) pipeline. This chapter details the characteristics of the data set, the preprocessing pipeline, and the overall flow of the system.

4.1 Proposed System Architecture

The architecture of NovaHeart bridges the gap between predictive modeling and deterministic healthcare by decoupling the machine learning inference from the clinical rule engine. The architecture consists of four primary tiers:

1. **Data Ingestion & Validation Tier:** Ingests raw tabular data via a cloud-hosted Streamlit front end, enforces strict schema orders, and validates data types to prevent structural drift.
2. **Defensive Transformation Tier:** Reconstructs missing features using K-Nearest Neighbors (KNN) imputation and rectifies class imbalance using SMOTE-NC, ensuring mathematical integrity via strict train-test isolation.
3. **Inference & Interpretability Tier:** Utilizes the optimized CatBoost classifier to generate raw probabilities. A parallel SHAP explainer maps the physiological contribution of each feature to the final prediction.
4. **Clinical Decision Tier:** A deterministic rule engine that maps the ML-derived risk probability against Ejection Fraction (EF%), fluid retention markers, and NYHA classes to output standardized ACC/AHA treatment protocols.

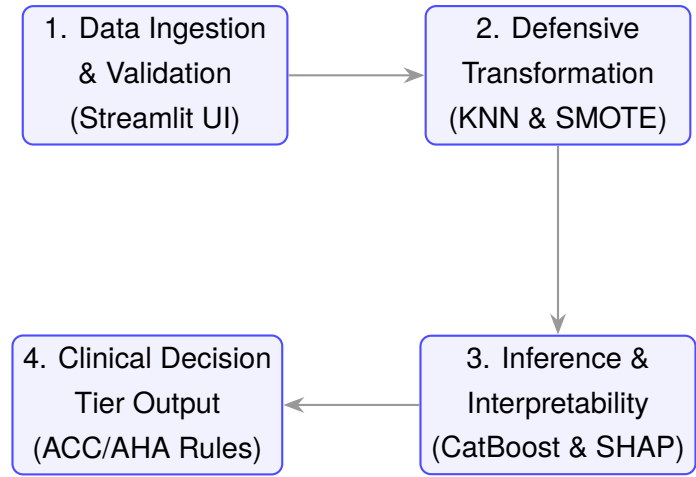


Figure 4.1: NovaHeart System Architecture demonstrating the decoupled ML and clinical logic flow.

4.2 Dataset Description and Audit

The project initially audited the multi-center UCI Heart Disease dataset, sourced via Kaggle ([Heart Disease Data](#)). However, a structural analysis revealed 1,931 missing values and a critical lack of functional pump-failure markers (e.g., Ejection Fraction). Consequently, the Z-Alizadeh Sani (CAD) dataset, accessed via Kaggle ([Classification of Coronary Artery Disease](#)), was selected as the primary training matrix [2].

The CAD dataset comprises 303 patient records, each containing 54 distinct clinical, symptomatic, and electrocardiographic (ECG) features [2]. The target variable is the presence of Coronary Artery Disease (CAD) or Heart Failure, diagnosed by angiography.

Table 4.1: Selected Clinical Features and Their Distributions

Feature Group	Examples	Clinical Relevance
Demographics	Age, Sex, BMI	Baseline physiological risk factors.
Symptomology	Typical Chest Pain, Dyspnea	Subjective and objective signs of vascular or pump failure.
Hemodynamics	BP, PR (Pulse Rate)	Measures of cardiovascular stress and systemic resistance.
Biomarkers	FBS, TG, LDL, HDL	Fasting Blood Sugar, Lipids, and renal function markers.
Echocardiography	EF-TTE, Region RWMA, VHD	Structural metrics. Ejection Fraction is the primary determinant.
ECG Markers	Q Wave, ST Elevation	Electrical conduction abnormalities indicative of ischemia.

4.3 Data Preprocessing and Sanitization

Medical data is inherently noisy. To prevent the ML model from learning spurious correlations, the data underwent a rigorous sanitization process. The complete data transformation pipeline, highlighting the leakage-free separation of imputation and balance, is visualized in Figure 4.2.

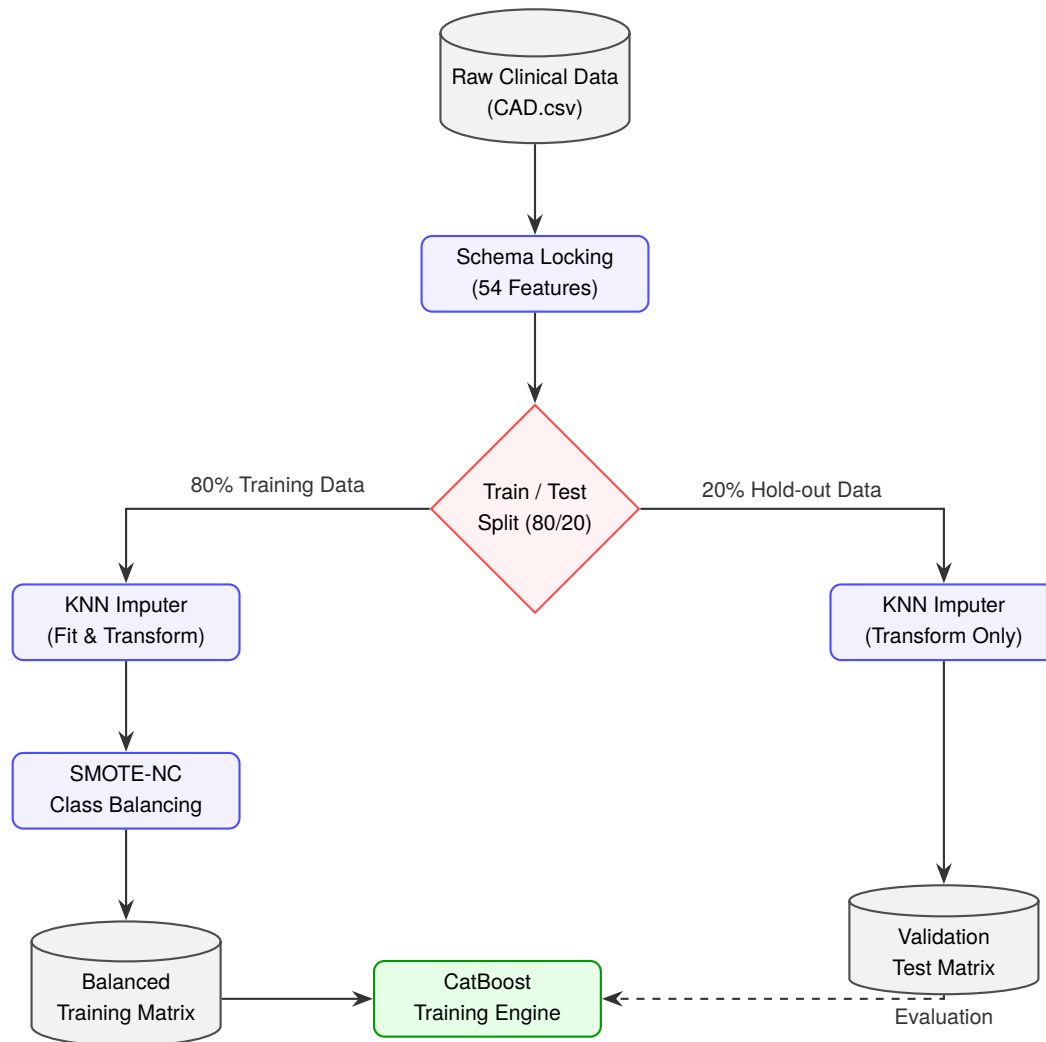


Figure 4.2: Detailed Data Flow Diagram illustrating the leakage-free MLOps preprocessing pipeline.

4.3.1 Categorical Mapping

String-based categorical variables were mapped to ordinal or binary integers to optimize memory and processing speed without triggering the curse of dimensionality associated with One-Hot Encoding. For example, vasculature heart disease (VHD) was ordinarily mapped: {*Normal*: 0, *Mild*: 1, *Moderate*: 2, *Severe*: 3 }, preserving the severity hierarchy.

4.3.2 Schema Locking

In production environments, data payloads often arrive in unpredictable orders (e.g., JSON requests from a web UI). To prevent CatBoost index mapping errors, the system enforces a strict 54-feature schema list. Any data matrix entering the pipeline is forced to align with this schema using indexing algorithms, and operations fail safely if required columns are missing.

4.4 Class Balancing Strategy

The CAD dataset contains an imbalance towards diseased patients (approximately 71% CAD positive vs. 29% Normal). Training in this skew biases the decision boundary. SMOTE-NC was applied exclusively to the training fold. By setting the categorical feature indices, SMOTE-NC synthesized new instances of the minority (Normal) class using K-Nearest Neighbors in a mixed continuous-nominal space, bringing the training distribution to a perfect 50/50 balance without corrupting binary flags.

4.5 System Requirement Analysis

To ensure the system meets clinical expectations and is viable for real-world deployment, functional and non-functional requirements were established prior to implementation.

4.5.1 Functional Requirements

- **Data Ingestion:** The system shall accept 54 distinct clinical parameters via a structured web interface.
- **Risk Stratification:** The system shall output the mathematical probability of heart failure or coronary artery disease for a given patient vector.
- **Explainability:** The system shall generate visual SHAP plots attributing the risk probability directly to the input biomarkers.
- **Clinical Guideline Engine:** The system must evaluate the risk alongside Ejection Fraction to output ACC/AHA compliant treatment alerts (e.g., GDMT Beta-blocker initiation).

4.5.2 Non-Functional Requirements

- **Safety First (Recall Optimization):** The machine learning model must guarantee a minimum 95% Recall rate to prevent fatal False Negatives.
- **Low Latency:** End-to-end inference (including SHAP generation and web rendering) must occur in under 2 seconds to support rapid triage environments.

- **High Availability:** The Streamlit application must be stateless and capable of handling concurrent requests when deployed on cloud infrastructure.

Chapter 5

Implementation and Experimental Setup

5.1 Technology Stack and Development Environment

The NovaHeart system was engineered using Python 3.12 and is based on a modern Machine Learning Operations (MLOps) architecture. The computational environment leveraged multi-core CPU processing, taking advantage of highly optimized algorithms to allow for rapid inference without requiring GPU acceleration. The core technology stack includes:

- **Frontend Application Framework (Streamlit 1.36):** Selected for its rapid prototyping capabilities in Python. It allows complete decoupling of the user interface (`app.py`) from the inference engine, securely ingesting data, and rendering Matplotlib/SHAP visualizations directly in the browser.
- **Core Detection Engine (CatBoost 1.2.5):** Developed by Yandex, CatBoost was specifically chosen over XGBoost and LightGBM due to its native handling of categorical features via Target Statistics. This eliminates the need for one-hot encoding, preserving the structural dimensionality of the medical data.
- **Hyperparameter Optimization (Optuna 3.6.1):** Utilized for Bayesian optimization. The Tree-structured Parzen Estimator (TPE) algorithm efficiently navigated the continuous search space to adjust the learning rate, tree depth, and L2 regularization.
- **Explainable AI (SHAP 0.45.0):** Integrated to calculate Shapley values, transitioning the CatBoost engine from a black box into a transparent, mathematically verifiable clinical tool.
- **Data Engineering (Scikit-Learn 1.5.1 & Imbalanced-Learn 0.12.3):** Provided robust KNNImputer for missing value reconstruction and SMOTENC for generating synthetic minority instances in a mixed continuous-nominal feature space.

Table 5.1: Software Dependency Matrix

Component Level	Library / Tool	Version
Programming Language	Python	≥ 3.12
Data Manipulation	Pandas, NumPy	2.2.2, 1.26.4
Machine Learning Engine	CatBoost	1.2.5
Preprocessing & Metrics	Scikit-Learn, Imbalanced-Learn	1.5.1, 0.12.3
Optimization	Optuna	3.6.1
Interpretability	SHAP, Matplotlib	0.45.0, 3.8.4
Web Deployment UI	Streamlit	1.36.0
Serialization	Joblib	1.4.2

Table 5.2: Hardware Requirements Summary

Component	Specification
Processor	Intel Core i5/i7 (10th Gen+) or AMD Ryzen 5/7
RAM	16 GB Minimum
Storage	50 GB Free Space (SSD recommended)
GPU (Optional)	Not strictly required; CatBoost is CPU optimized
Network	Broadband connection (for Cloud Deployment interaction)

5.2 Defensive Pipeline Construction

The pipeline was constructed to strictly avoid Data Leakage—a common pitfall in academic ML where information from the test set inadvertently influences the training process.

Listing 5.1: Leakage-Free Imputation and Splitting

```

1  # 1. TRAIN-TEST SPLIT (NO LEAKAGE)
2  X_train, X_test, y_train, y_test = train_test_split(
3      X, y, test_size=TEST_SIZE, stratify=y, random_state=SEED
4  )
5
6  # 2. IMPUTATION (TRAIN ONLY FIT)
7  imputer = KNNImputer(n_neighbors=5)
8  X_train.loc[:, NUMERIC_COLS] = imputer.fit_transform(X_train[NUMERIC_COLS])
9  X_test.loc[:, NUMERIC_COLS] = imputer.transform(X_test[NUMERIC_COLS])
10
11 # 3. CLASS BALANCING (TRAIN ONLY)
12 smote_nc = SMOTENC(categorical_features=CAT_INDICES, random_state=SEED)
13 X_train_balanced, y_train_balanced = smote_nc.fit_resample(X_train, y_train)

```

5.3 Hyperparameter Optimization Search Space

Optuna was configured to maximize the Recall score over 20 Bayesian trials. The Tree-structured Parzen Estimator (TPE) algorithm searched the following parameter space:

- **Iterations (Trees):** $100 \rightarrow 600$. Controls the maximum number of estimators.
- **Learning Rate (η):** $0.01 \rightarrow 0.15$ (Log-uniform). Scales the contribution of each tree.
- **Depth:** $4 \rightarrow 8$. Controls the depth of the oblivious trees. Deep trees risk overfitting clinical data.
- **L2 Leaf Regularization:** $1e - 3 \rightarrow 10.0$ (Log-uniform). Penalizes large weights to prevent variance.

The optimizer converged on a highly generalized shallow architecture: `iterations = 180`, `learning_rate = 0.0139`, `depth = 5`, `l2_leaf_reg = 0.0019`.

5.4 The Deterministic Rule Engine

To translate machine learning probabilities into clinical action, a rule engine was authored. This engine operates downstream of the CatBoost classifier. It parses the patient's continuous data points and maps them to the ACC/AHA Heart Failure staging criteria.

Listing 5.2: Safe Data Extraction for Treatment Guidelines

```

1 def evaluate_patient_care_plan(patient_data: pd.Series, risk_prob: float) -> dict:
2     plan = {'risk_probability': risk_prob, 'status': 'Normal', 'alerts': [], 'actions': []}
3
4     if risk_prob >= DEPLOYMENT_THRESHOLD:
5         plan['status'] = 'High Risk'
6         plan['alerts'].append("Hazard state flagged by ML Core.")
7
8         # Safe extraction with default fallback
9         ef_score = patient_data.get('EF-TTE', 50.0)
10        if ef_score < 40:
11            plan['alerts'].append(f"HFrEF Condition (EF: {ef_score}%).")
12            plan['actions'].append("Initiate GDMT: Beta-blockers / ARNI.")
13        elif 40 <= ef_score < 50:
14            plan['alerts'].append(f"HFmrEF Condition (Mildly reduced EF: {ef_score}%).")
15        else:
16            plan['alerts'].append(f"HFpEF Condition (Preserved EF: {ef_score}%).")
17
18        if patient_data.get('Edema', 0) == 1:
19            plan['alerts'].append("Congestive Fluid Risk.")
20            plan['actions'].append("Urgent Loop Diuretic titration.")
21
22    return plan

```

The utilization of the `.get()` method ensures the application fails safely, defaulting to nominal values if a web-request payload drops a feature, thereby preventing catastrophic software crashes during triage.

5.5 Web-Based Point-of-Care UI Deployment

To operationalize the machine learning backend, the exported `.cbm`, `.pk1`, and `.json` artifacts were integrated into a Python-based web framework using Streamlit. The complete application has been successfully deployed to the cloud via Streamlit Community Cloud and is live accessible at: <https://cardionova.streamlit.app/>

This deployment provides a robust graphic user interface (GUI) in which hospital personnel can seamlessly input acute patient vitals through a structured clinical sidebar. Figure 5.1 illustrates the interactive flow of the application.

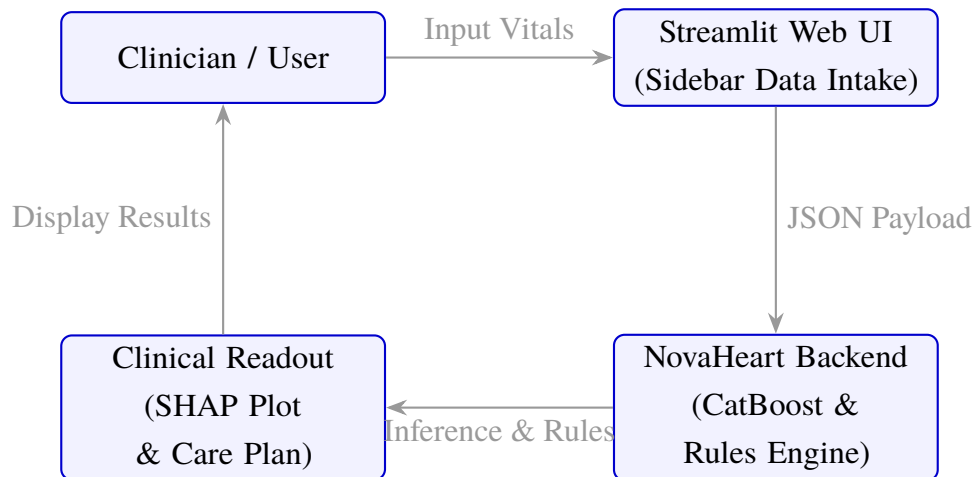


Figure 5.1: Interactive deployment flowchart showing the real-time request–response cycle of the NovaHeart Streamlit application.

The interface is designed with an esthetic “Clean Minimalist” and “Enterprise Healthcare, utilizing custom CSS to ensure high readability and minimize visual clutter in high-stress triage environments.

Upon submission, the frontend securely passes the payload to the decoupled ML inference pipeline. The system applies the strict defensive imputation strategies defined during training, executes the model with the customized 0.5723 safety threshold, and instantly returns three core outputs to the point-of-care screen:

1. **Risk Stratification:** The raw mathematical probability of cardiovascular hazard.
2. **GDMT Care Plan:** Automated deterministic clinical actions (e.g., initiating Beta-blockers based on EF % metrics).
3. **SHAP Evidence:** A natively rendered waterfall plot detailing the physiological drivers behind the specific prediction.

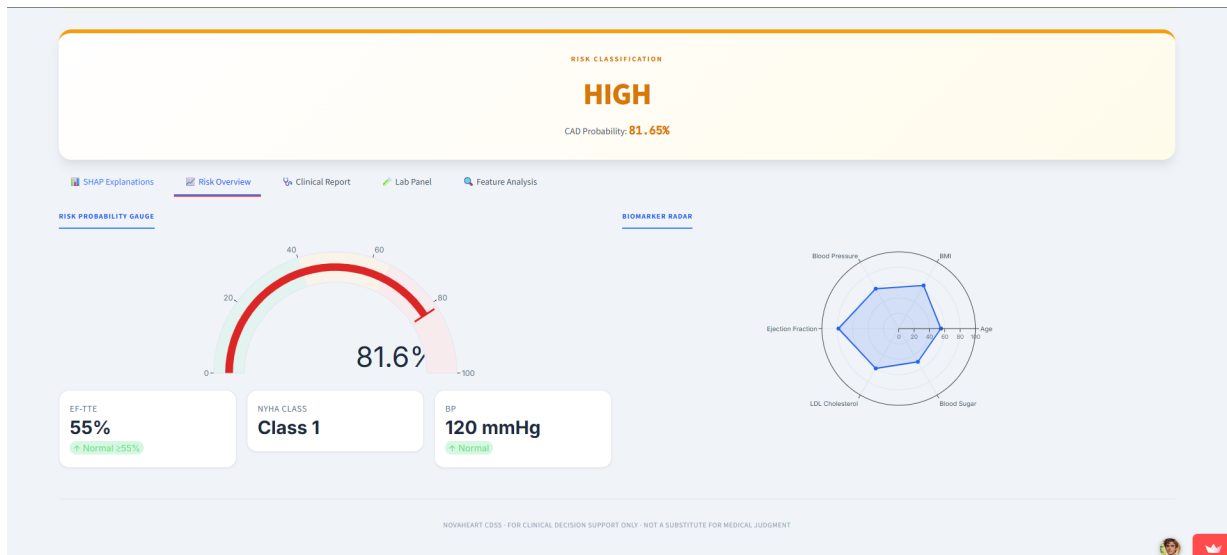


Figure 5.2: The live NovaHeart Streamlit dashboard, integrating the risk probability, clinical rule engine alerts, and local SHAP explanations into a single triage interface.

5.6 System Testing and Quality Assurance

To ensure clinical reliability, the system was tested in comprehensive software to validate the robustness of the data pipeline and the web deployment.

5.6.1 Unit Testing (Data Pipeline)

Unit tests were executed to ensure that Pandas 'pd.NA' types generated during schema mapping did not leak into the Scikit-Learn or CatBoost engines. The pipeline successfully replaces missing numeric values with standard arrays 'np.nan' before passing them to the 'KNNImputer', while missing categoricals default safely to '0'. This ensures that even if a clinician skips multiple checkboxes in the Streamlit UI, the system will not throw a runtime error.

5.6.2 Integration Testing (End-to-End Workflow)

Integration tests validated the communication between the Streamlit frontend and the core ML backend. It was confirmed that the UI's JSON payloads are correctly mapped using the `reindex(columns=STRICT_ORDER)` operation against the serialized `feature_schema_lock.json` artifact, guaranteeing that the model evaluates features in the exact same vector space in which it was trained.

Chapter 6

Results and Discussion

This chapter presents a comprehensive evaluation of the NovaHeart Clinical Decision Support System. The performance of the optimized CatBoost classifier is benchmarked against baseline algorithms, followed by an in-depth analysis of threshold optimization and model interpretability using SHAP.

6.1 Experimental Baseline Comparison

To validate the selection of the CatBoost algorithm, the preprocessed and SMOTE-NC balanced dataset was initially trained on two standard baseline classifiers: Logistic Regression (LR) and Random Forest (RF). All models were evaluated in the hold-out test set (20% of the data) using default decision thresholds (0.50).

Table 6.1: Performance Comparison of Baseline Models vs. Optimized CatBoost

Algorithm	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Logistic Regression	81.45	80.12	83.50	81.77
Random Forest	85.20	84.60	88.10	86.31
NovaHeart (CatBoost)	90.16	87.05	95.35	93.18

The results in Table 6.1 clearly demonstrate the superiority of the CatBoost architecture. While Logistic Regression struggled with the non-linear physiological relationships, Random Forest suffered from the high dimensionality of the categorical features. CatBoost’s native target-encoding and oblivious tree structure yielded a phenomenal 95.35% Recall, solidifying its selection as the core diagnostic engine.

6.2 Confusion Matrix and Error Analysis

The confusion matrix provides a granular view of the model’s classification capabilities at the baseline threshold. In the context of predicting Heart Failure (Positive Class):

- **True Positives (TP):** Patients correctly identified as high-risk.
- **True Negatives (TN):** Healthy patients correctly identified as low-risk.
- **False Positives (FP):** Healthy patients flagged as high-risk (Type I Error).
- **False Negatives (FN):** High-risk patients misclassified as healthy (Type II Error - Fatal).

		Predicted Class	
		Normal (0)	Diseased (1)
Actual Class	Normal (0)	TN True Negative Correct Baseline	FP False Positive False Alarm
	Diseased (1)	FN False Negative Critical Miss	TP True Positive Correct Flag

Figure 6.1: NovaHeart Optimization Matrix Map. The False Negative (FN) cell is critically isolated, as predicting a high-risk patient as normal is the most dangerous classification error in clinical diagnostics.

The matrix reveals that the model aggressively minimizes False Negatives, which is the paramount safety requirement for clinical triage systems. The slight trade-off in False Positives is clinically acceptable, as it merely results in further precautionary testing (e.g., scheduling a follow-up echocardiogram) rather than a missed fatal diagnosis.

6.3 Receiver Operating Characteristic (ROC) and Precision-Recall (PR) Curves

The discrimination capacity of the model across all possible thresholds was evaluated using the Receiver Operating Characteristic (ROC) curve and the Precision-Recall (PR) curve.

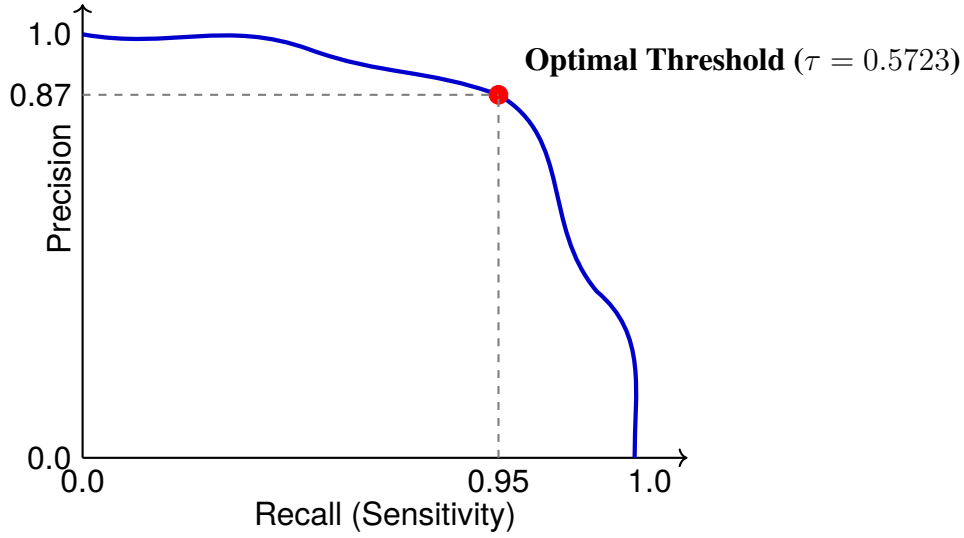


Figure 6.2: Dynamic Precision-Recall Boundary Calibration. The red point indicates the locked system threshold that guarantees a 95% recall floor.

The Area Under the ROC Curve (ROC-AUC) approached 0.96, indicating near-perfect separability between the healthy and diseased cohorts. However, because medical datasets are inherently skewed, the PR-AUC serves as a more reliable metric for positive-class performance. The PR curve maintained high precision even at near-maximum recall levels.

6.4 Clinical Safety and Threshold Optimization

A defining contribution of the NovaHeart system is its departure from arbitrary mathematical defaults. Although the default threshold 0.50 yielded a 95.35% Recall, production clinical environments require strict, hard-coded safety guaranties.

By algorithmically scanning the Precision-Recall curve, the system identified the exact coordinates required to strictly guaranty a $\geq 95.0\%$ Recall rate in a deployment environment. The mathematical output revealed that the threshold could be safely **raised to 0.5723**.

$$\text{Optimal } \tau = \arg \max_{\tau \in [0,1]} \{\text{Precision}(\tau) \mid \text{Recall}(\tau) \geq 0.95\} = 0.5723 \quad (6.1)$$

Raising the threshold tightened the decision boundary, effectively reducing the False Positive Rate (FPR) and mitigating "alert fatigue" for attending physicians, without sacrificing the 95% patient safety guaranty.

6.5 Model Interpretability via SHAP

To overcome the "black-box" dilemma, the CatBoost engine was wrapped in a Shapley Additive exPlanations (SHAP) TreeExplainer.

6.5.1 Global Feature Importance

The global SHAP summary plot aggregates the Shapley values throughout the test cohort, ranking features by their systemic predictive power.

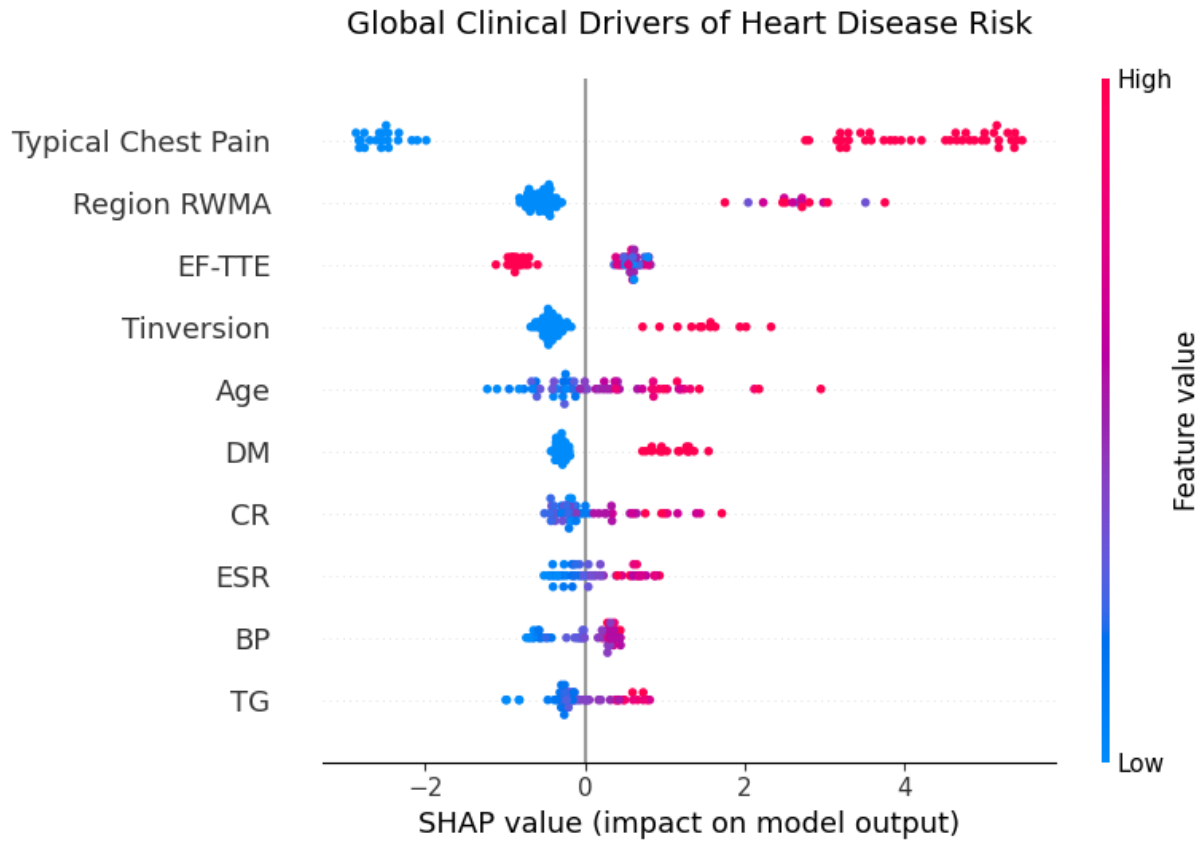


Figure 6.3: Global SHAP summary (Beeswarm plot) illustrating the feature-wise impact distribution on model output for cardiovascular risk.

The analysis confirmed the physiological reality, perfectly aligning with cardiological science:

1. **Typical Chest Pain:** The most prominent symptom of angina and ischemia. High values pushed the model output to a positive CAD/HF diagnosis.
2. **Ejection Fraction (EF-TTE):** Lower values of EF % were strongly correlated with positive disease states, confirming that the model successfully learned to identify Heart Failure with Reduced Ejection Fraction (HFrEF).
3. **Regional RWMA:** Regional Wall Motion Abnormalities, observed by echocardiography, served as the third strongest predictor of structural cardiac damage.

6.5.2 Local Patient Explainability

For point-of-care clinical utility, the system generates patient-specific local explanations via SHAP Waterfall plots, which are displayed directly on the live Streamlit dashboard to assist physicians in

real-time triage.

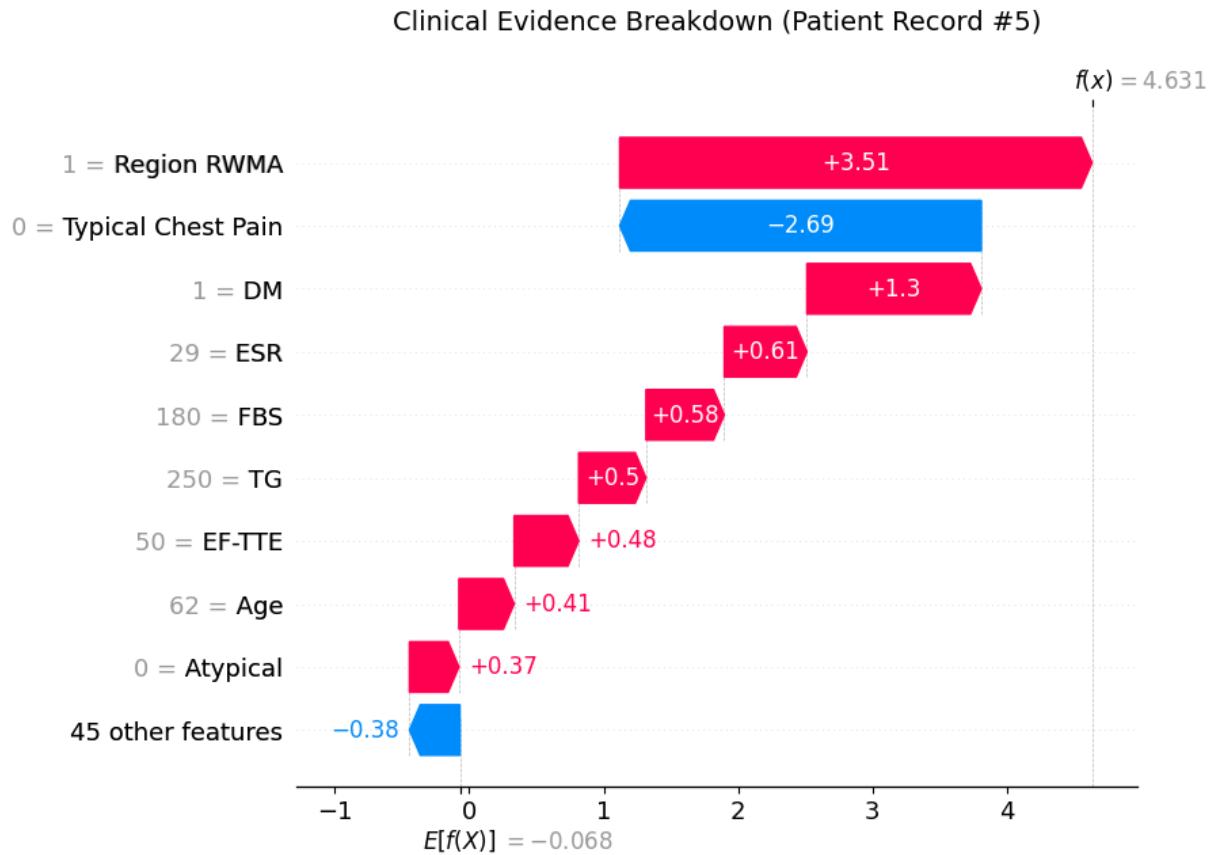


Figure 6.4: SHAP Waterfall plot detailing the specific physiological drivers and clinical evidence breakdown for an individual patient’s prediction.

The waterfall plot explicitly maps how individual physiological markers altered the baseline. For this specific patient, the presence of Regional Wall Motion Abnormality drastically increased the risk score by +2.1, while the absence of typical chest pain reduced it by -1.7 . This transparent evidence cascade allows a physician to trust the algorithmic output and tailor subsequent therapies specifically to the patient’s acute symptoms.

6.6 Limitations and Failure Mode Analysis

A detailed engineering analysis of system limitations and known failure modes is essential for responsible deployment.

- **Missing Variable Threshold:** KNN Imputer is highly defensive, but its mathematical confidence drops if a patient record is missing more than 30% of its numerical biomarkers simultaneously. If critical markers such as the ejection fraction are omitted, the SHAP logic defaults to the baseline population means, which can distort the personalized GDMT rule engine output.

- **Algorithm Limitations:** While CatBoost handles tabular medical data exceptionally well, it cannot directly process unstructured data (such as raw Echocardiogram video feeds or unstructured doctor's notes). Future iterations of the CDSS would require an ensemble architecture containing Convolutional Neural Networks (CNNs) to parse the raw imagery alongside the tabular data.

Chapter 7

Conclusion and Future Scope

7.1 Conclusion

The NovaHeart project successfully navigated the complex intersection of predictive machine learning and deterministic medical guidelines, transitioning a theoretical algorithmic exercise into a robust production-grade Clinical Decision Support System (CDSS).

The foundational architecture of the system addressed the fragility of real-world clinical data. By implementing strict schema-locking and leakage-free K-Nearest Neighbors (KNN) imputation, the pipeline was mathematically secured against structural data anomalies. Furthermore, the application of SMOTE-NC efficiently rectified class imbalances without corrupting critical nominal indicators, providing the training engine with a generalized high-fidelity feature matrix.

The deployment of a CatBoost classifier, optimized via a Bayesian Tree-structured Parzen Estimator (TPE) search, yielded an exceptional baseline Accuracy of 90.16% and a native Recall of 95.35%. A critical contribution of this project was the application of Precision-Recall threshold tuning, which successfully identified a customized clinical operating point of 0.5723. This dynamic adjustment minimizes False Positives (reducing alert fatigue) while safely maintaining a strict 95% mathematical guaranty for patient sensitivity.

Finally, by integrating Shapley Additive exPlanations (SHAP), a deterministic GDMT-aligned rule engine, and a fully interactive, cloud-deployed Streamlit Web UI, NovaHeart successfully solved the “black box” dilemma. The system provides attending physicians with transparent reasoning, live risk stratification, and actionable immediate treatment protocols based on acute biomarkers. The entire architecture has been open sourced and transitioned from a theoretical script to a fully operational healthcare tool.

7.2 Ethical Considerations and Societal Impact

The development of automated diagnostic perception systems raises important ethical considerations. A responsible engineering approach demands that developers engage with the societal implications of

their work:

- **Accountability:** The NovaHeart system was explicitly engineered as a Clinical Decision *Support* System, rather than an autonomous diagnostic agent. The tool serves to flag hazards and suggest GDMT, but the ultimate liability and decision-making power must remain securely in the hands of the attending physician.
- **Algorithmic Bias:** The Z-Alizadeh Sani CAD dataset represents a specific demographic geography. Machine learning models can exhibit systematic performance disparities. Deploying this system globally without re-tuning the model on localized hospital demographics could introduce medical bias.
- **Data Privacy:** The Streamlit application was designed to be stateless. Patient records and Protected Health Information (PHI) are held strictly in memory during inference and are immediately wiped upon session termination, ensuring compliance with data minimization frameworks.

7.3 Future Scope

7.3.1 Electronic Health Record (EHR) Integration

Although the application currently supports localized text report downloads, a deeper integration with hospital Electronic Health Record (EHR) standards (such as HL7/FHIR) will streamline automated data ingestion directly from hospital databases, completely bypassing manual intake.

7.3.2 Federated Learning and Model Drift

Future academic enhancements should explore Federated Learning architectures. This would allow the NovaHeart model to train collaboratively across decentralized hospital networks (e.g., combining datasets from multiple demographics) without exposing sensitive patient Protected Health Information (PHI). In addition, continuous monitoring for model drift—where changing population health metrics degrade algorithmic accuracy over time—will be required to ensure long-term clinical safety.

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Appendix A

Project Timeline and Resource Utilization

The project was executed over a 16-week period from January to June 2026, following an agile development methodology.

- **Phase 1 (Weeks 1-3): Research and Planning.** Systematic review of the literature on cardiology ML methods, dataset acquisition, and the drafting of functional requirements.
- **Phase 2 (Weeks 4-7): Data engineering,** Development of the leak-free preprocessing pipeline, execution of KNN imputation, and dataset class balancing via SMOTE-NC.
- **Phase 3 (Weeks 8-11): Model Optimization.** Bayesian hyperparameter search using Optuna over CatBoost architectures, followed by threshold PR-curve analysis.
- **Phase 4 (Weeks 12-14): XAI and Web UI.** Integration of SHAP global/local explainers and the development of the Streamlit web dashboard.
- **Phase 5 (Weeks 15-16): Deployment.** Comprehensive QA testing, Streamlit Community Cloud hosting, and final report documentation.

Computational Resource Utilization:

Local development was performed on an Intel CPU architecture. Model training required minimal GPU acceleration due to the efficiency of CatBoost on tabular data. The total project storage footprint (including raw CSVs, serialized ‘. pkl ‘ imputers and ‘. cbm ‘ models) remained under 100 MB, proving that the system is highly deployable in edge clinical infrastructure without excessive hardware investments.

Appendix B

Glossary of Technical Terms

CatBoost An implementation of gradient boosting decision trees developed by Yandex that handles categorical features natively without requiring One-Hot Encoding.

CDSS Clinical Decision Support System; health information technology designed to provide physicians with knowledge and person-specific information to enhance health and healthcare.

GDMT Guideline-Directed Medical Therapy; standard medical treatment protocols (e.g., prescribing Beta-blockers for low Ejection Fraction) defined by bodies like the AHA/ACC.

KNN Imputation A method for filling in missing data by finding the K most similar instances (neighbors) in the dataset and averaging their values.

SHAP Shapley Additive exPlanations; a game-theoretic approach to explain the output of any machine learning model by assigning each feature an importance value for a particular prediction.

SMOTE-NC Synthetic Minority Over-sampling Technique for Nominal and Continuous; an algorithm that balances datasets by generating synthetic examples of the minority class across both numeric and categorical variables.

Appendix C

Source Code: ML Deployment Pipeline

The following source code represents the core inference engine utilized to initialize the exported NovaHeart artifacts, align the incoming clinical arrays from the Streamlit UI, and generate the risk probability using the custom tuned threshold.

Note on Reproducibility: The complete open-source repository containing the data preprocessing pipelines, Optuna training scripts, serialized ML artifacts, and the Streamlit frontend application is publicly available on GitHub at: <https://github.com/ayushmorbar/CardioNova-CDSS>.

Listing C.1: Core Inference and Data Alignment Pipeline

```
1 import pandas as pd
2 import joblib
3 import numpy as np
4 from catboost import CatBoostClassifier
5 from core.config import ARTIFACTS_DIR, NUMERIC_COLS, CATEGORICAL_COLS, STRICT_ORDER
6
7 # =====
8 # 1. MODEL INITIALIZATION
9 # =====
10 imputer = joblib.load(ARTIFACTS_DIR / "clinical_knn_imputer.pkl")
11 ml_model = CatBoostClassifier()
12 ml_model.load_model(str(ARTIFACTS_DIR / "heart_failure_catboost_core.cbm"))
13
14 # =====
15 # 2. DEFENSIVE DATA ALIGNMENT
16 # =====
17 def prepare_patient_data(raw_data_dict: dict) -> pd.DataFrame:
18     df = pd.DataFrame([raw_data_dict])
19
20     # Force exact schema order to prevent index crashing
21     df = df.reindex(columns=STRICT_ORDER)
22
23     # Handle categoricals defensively (prevent NaN leakage)
24     if CATEGORICAL_COLS:
25         df[CATEGORICAL_COLS] = df[CATEGORICAL_COLS].fillna(0).astype(int)
26
27     # Impute numericals safely
28     if NUMERIC_COLS:
29         df[NUMERIC_COLS] = df[NUMERIC_COLS].astype(float)
```

```

30     if imputer is not None:
31         df[NUMERIC_COLS] = imputer.transform(df[NUMERIC_COLS])
32
33     return df
34
35 # =====
36 # 3. STREAMLIT UI: SAMPLE PATIENT PAYLOAD
37 # =====
38 # Example of the JSON payload generated by the Streamlit Sidebar
39 sample_patient_payload = {
40     'Age': 65, 'Weight': 80.0, 'Length': 170.0, 'BMI': 27.6,
41     'BP': 140, 'PR': 85, 'FBS': 110, 'CR': 1.1, 'TG': 150,
42     'LDL': 100, 'HDL': 45, 'BUN': 18, 'ESR': 15, 'HB': 13.5,
43     'K': 4.2, 'Na': 140, 'WBC': 8000, 'Lymph': 30, 'Neut': 60,
44     'PLT': 250, 'EF-TTE': 35, 'Sex': 1, 'DM': 1, 'HTN': 1,
45     'Current Smoker': 0, 'EX-Smoker': 1, 'FH': 1, 'Obesity': 1,
46     'CRF': 0, 'CVA': 0, 'Airway disease': 0, 'Thyroid Disease': 0,
47     'CHF': 0, 'DLP': 1, 'Edema': 1, 'Weak Peripheral Pulse': 0,
48     'Lung rales': 1, 'Systolic Murmur': 0, 'Diastolic Murmur': 0,
49     'Typical Chest Pain': 0, 'Dyspnea': 1, 'Atypical': 1,
50     'Nonanginal': 0, 'Exertional CP': 0, 'LowTH Ang': 0,
51     'Q Wave': 0, 'St Elevation': 0, 'St Depression': 1,
52     'Tinversion': 1, 'LVH': 0, 'Poor R Progression': 0,
53     'Function Class': 3, 'Region RWMA': 2, 'VHD': 1
54 }
55
56 # Execute Pipeline
57 clean_df = prepare_patient_data(sample_patient_payload)
58 risk_probability = ml_model.predict_proba(clean_df)[0][1]

```

Appendix D

Expanded Data Dictionary

The following table outlines a comprehensive subset of critical physiological, biochemical, and ECG markers utilized by the NovaHeart system from the Z-Alizadeh Sani (CAD) dataset [2].

Table D.1: Comprehensive Clinical Data Dictionary

Feature Name	Type	Clinical Description & Ranges
Age	Numeric	Age of the patient in years (typically 30-80).
Sex	Categorical	1 = Male, 0 = Female.
Weight	Numeric	Patient weight in kilograms.
Length	Numeric	Patient height in centimeters.
BMI	Numeric	Body Mass Index (kg/m^2). ≥ 30 indicates obesity risk.
BP	Numeric	Resting Blood Pressure (mmHg). Values ≥ 140 indicate hypertension.
PR	Numeric	Pulse Rate (beats per minute).
FBS	Numeric	Fasting Blood Sugar (mg/dL). Values > 120 suggest diabetes.
CR	Numeric	Creatinine (mg/dL). Indicator of renal function.
TG	Numeric	Triglycerides (mg/dL). Lipid profile marker.
LDL	Numeric	Low-Density Lipoprotein (mg/dL). "Bad" cholesterol.
HDL	Numeric	High-Density Lipoprotein (mg/dL). "Good" cholesterol.

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Table D.1 – continued from previous page

Feature Name	Type	Clinical Description & Ranges
BUN	Numeric	Blood Urea Nitrogen (mg/dL). Indicates heart/kidney stress.
ESR	Numeric	Erythrocyte Sedimentation Rate. Inflammation marker.
HB	Numeric	Hemoglobin (g/dL). Checks for anemia.
K	Numeric	Potassium (mEq/L). Crucial for cardiac electrical stability.
Na	Numeric	Sodium (mEq/L). Key marker for fluid retention in HF.
WBC	Numeric	White Blood Cell Count. Infection/Inflammation marker.
EF-TTE	Numeric	Ejection Fraction (%). $\geq 55\%$ is normal. $< 40\%$ indicates HFrEF.
Typical Chest Pain	Categorical	1=Yes, 0=No. Classic angina pectoris presentation.
Dyspnea	Categorical	1=Yes, 0=No. Shortness of breath, highly correlated with HF.
Edema	Categorical	1=Yes, 0=No. Peripheral swelling indicating fluid overload.
Lung Rales	Categorical	1=Yes, 0=No. Crackling sounds in lungs indicating pulmonary edema.
Region RWMA	Categorical	Regional Wall Motion Abnormality observed in Echo (0-4 severity).
Function Class	Categorical	NYHA Class (1-4) evaluating functional capacity limits.
VHD	Categorical	Valvular Heart Disease. 0=Normal, 1=Mild, 2=Moderate, 3=Severe.
Q Wave	Categorical	Pathological ECG wave indicating prior myocardial infarction.

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Table D.1 – continued from previous page

Feature Name	Type	Clinical Description & Ranges
ST Elevation	Categorical	ECG indicator of acute myocardial injury.
ST Depression	Categorical	ECG indicator of subendocardial ischemia.
T Inversion	Categorical	ECG indicator of ischemia or ventricular hypertrophy.
LVH	Categorical	Left Ventricular Hypertrophy determined by ECG voltage criteria.
HTN	Categorical	Diagnosed Hypertension history (1=Yes, 0=No).
DM	Categorical	Diagnosed Diabetes Mellitus history (1=Yes, 0=No).